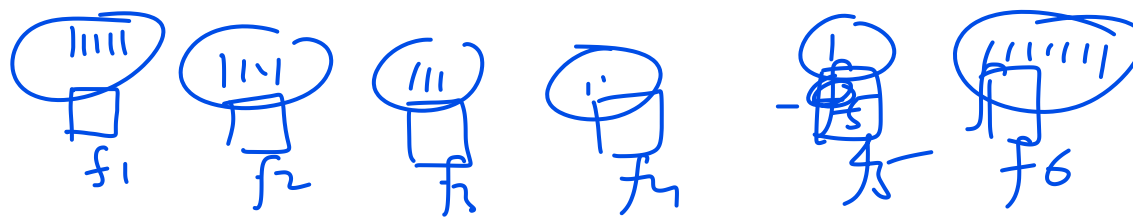


Lec-0 1, 2, 3, 4, 8 30-40

Lec_63

4.) Optimal merge Pattern



Merging small files into big files with minimum number of moves

- Algorithm is same as Huffman Coding Algorithm (min heap)
 - Using min heap $TC = O(n \log n)$
- $TC = O(n \log n)$

Again I am saying 14, greedy

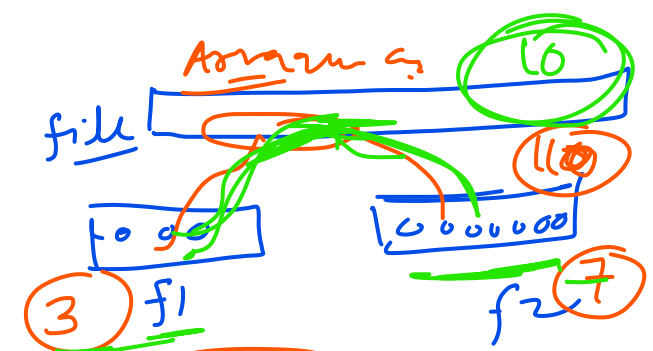
① Numerical 1.

② TC 1.

③ MST (various)

Ex:

- 1) a=20, b=35, c=10, d=18, e=17, f=15, g=22 find
- a) Min number of moves while merging the all files
 - b) min number of Comparisons
 - c) max number of comparisons



~~a=20~~, ~~b=35~~, ~~c=10~~, ~~d=18~~, ~~e=17~~, ~~f=15~~, ~~g=22~~, ~~t1=25~~

~~t1=77~~, ~~t2=35~~, ~~t3=42~~

a) $25 + 60 + 35 + 42 + 77 + 137$

$t1 = 77$, $t2 = 35$, $t3 = 42$

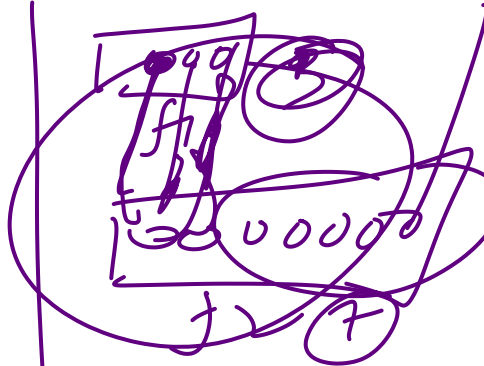
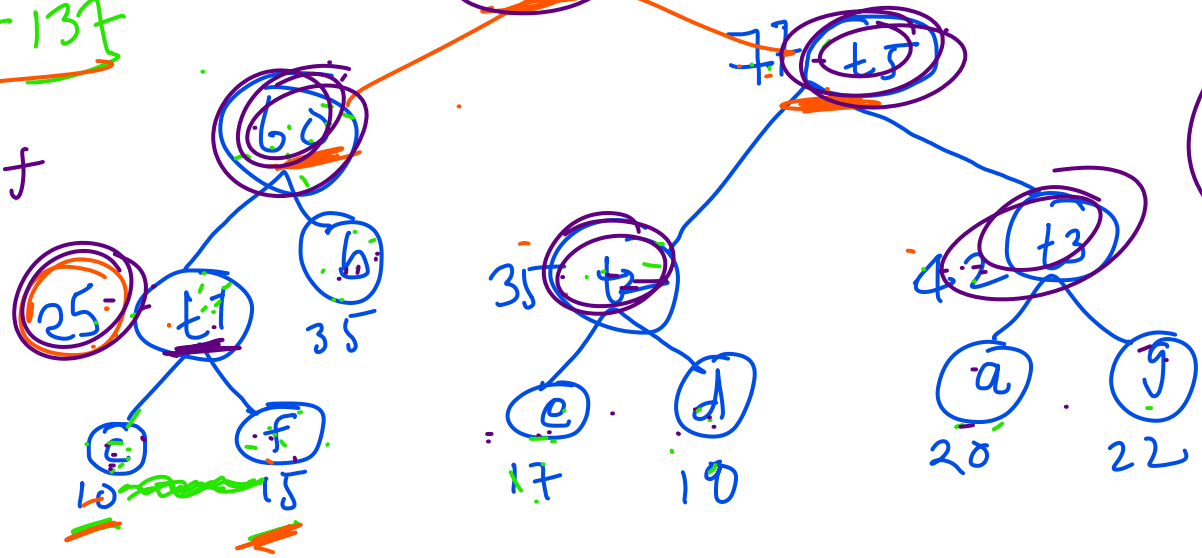
moves

m n

more = $m+n$ each

Ans

q) moves = sum of all internal nodes of tree



⑥ min no. of ~~comp~~^{ops} $\Rightarrow \min(m, n)$
 $10 + 17 + 20 + 25 + 35 + 60$ (can't ~~sleep~~)
 some of min of each siblings

⑦ max no. of ~~comp~~^{ops} $\Rightarrow \underline{m+n-1}$
 $25-1 + 35-1 + 42-1 + 60-1 + 77-1 + 137-1$
 Can't do

$\Rightarrow$ Sum of Internal nodes - no. of Internal
 $\underline{25 + 35 + 42 + 60 + 77 + 137 - 6}$

Can't do all these
Answer

mst

